

LAB EXPERIMENT RECORD

Set 14: Survey Results Analysis

Department	B.Tech AIDS
Course Code / Course Name	DSA06 / Data Handling and Visualization
Experiment	Data Visualization using R
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Aim

To analyze survey results using a stacked bar chart, radar chart, and table in R, and to build an interactive R Shiny dashboard combining all three.

Dataset: Survey Results

Respondent	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

PROGRAM 1: Stacked Bar Chart for Question 1

Create a stacked bar chart to display the distribution of answers for Question 1. Label the chart elements.

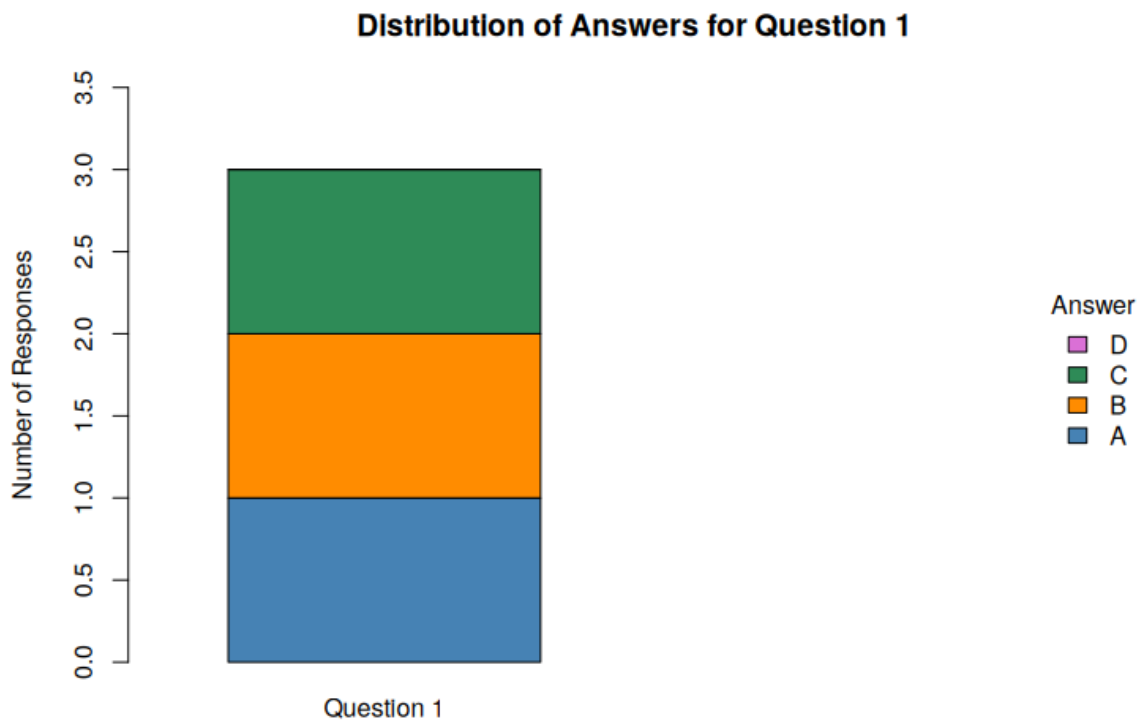
```
# Create dataset for Set 14
survey_results <- data.frame(
  Respondent = c(1, 2, 3),
  Question_1 = c("A", "B", "C"),
  Question_2 = c("B", "A", "A"),
  Question_3 = c("C", "D", "B")
)

print(survey_results)

# Count answers for Question 1 (all answer options A to D)
q1_counts <- table(factor(survey_results$Question_1, levels = c("A", "B", "C", "D")))
q1_matrix <- matrix(q1_counts, ncol = 1,
  dimnames = list(names(q1_counts), "Question 1"))
print(q1_matrix)

# 1. Stacked Bar Chart for Question 1
barplot(q1_matrix,
  col = c("steelblue", "darkorange", "seagreen", "orchid"),
  width = 0.5, xlim = c(0, 1.5),
  ylab = "Number of Responses",
  main = "Distribution of Answers for Question 1",
```

```
ylim = c(0, 3.5),  
legend.text = TRUE,  
args.legend = list(title = "Answer", x = "right", bty = "n"))
```



Output 1: Distribution of Answers for Question 1

Observation: The single stacked bar for Question 1 is divided equally into three parts: options A, B, and C each received one response, and option D received none. The responses for Question 1 are evenly split.

PROGRAM 2: Radar Chart of Response Patterns

Generate a radar chart to visualize the overall pattern of responses across all three questions.

```
# Create dataset for Set 14
survey_results <- data.frame(
  Respondent = c(1, 2, 3),
  Question_1 = c("A", "B", "C"),
  Question_2 = c("B", "A", "A"),
  Question_3 = c("C", "D", "B")
)

print(survey_results)

# Answer counts for each question
options <- c("A", "B", "C", "D")
counts <- rbind(
  "Question 1" = table(factor(survey_results$Question_1, levels = options)),
  "Question 2" = table(factor(survey_results$Question_2, levels = options)),
  "Question 3" = table(factor(survey_results$Question_3, levels = options))
)
print(counts)

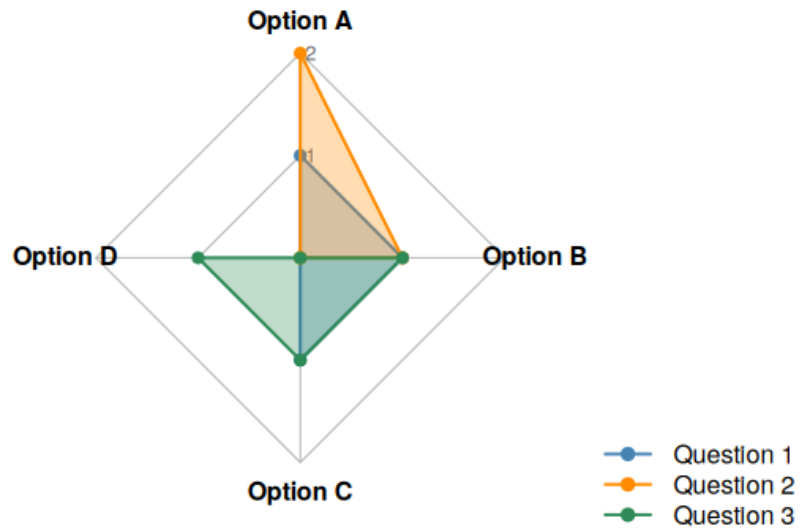
# 2. Radar Chart (base R)
n_axes <- length(options)
angles <- pi / 2 - 2 * pi * (0:(n_axes - 1)) / n_axes
max_val <- 2

plot(NA, xlim = c(-1.3, 1.3), ylim = c(-1.3, 1.3), asp = 1, axes = FALSE,
     xlab = "", ylab = "", main = "Radar Chart of Survey Response Patterns")

# Grid rings and axis lines
for (r in 1:max_val) {
  polygon(r / max_val * cos(angles), r / max_val * sin(angles), border = "gray70")
  text(0.05, r / max_val, labels = r, cex = 0.8, col = "gray40")
}
segments(0, 0, cos(angles), sin(angles), col = "gray70")
text(1.15 * cos(angles), 1.15 * sin(angles), labels = paste("Option", options), font = 2)

# One polygon per question
q_cols <- c("steelblue", "darkorange", "seagreen")
for (i in 1:nrow(counts)) {
  r <- counts[i, ] / max_val
  polygon(r * cos(angles), r * sin(angles),
          col = adjustcolor(q_cols[i], alpha.f = 0.3), border = q_cols[i], lwd = 2)
  points(r * cos(angles), r * sin(angles), pch = 19, col = q_cols[i])
}
legend("bottomright", legend = rownames(counts), col = q_cols,
      lwd = 2, pch = 19, bty = "n")
```

Radar Chart of Survey Response Patterns



Output 2: Radar Chart of Survey Response Patterns

Observation: Each question forms a different shape on the radar chart. Question 2 has a sharp peak towards Option A (2 responses), showing the strongest agreement among respondents. Question 1 spreads across A, B, and C, while Question 3 spreads across B, C, and D. Option B is the only answer chosen in all three questions.

PROGRAM 3: Table of Survey Results

Build a table to show the survey response data for each respondent. Label the table elements.

```
# Create dataset for Set 14
survey_results <- data.frame(
  Respondent = c(1, 2, 3),
  Question_1 = c("A", "B", "C"),
  Question_2 = c("B", "A", "A"),
  Question_3 = c("C", "D", "B")
)

print(survey_results)

# 3. Table of survey results
results_table <- survey_results
colnames(results_table) <- c("Respondent", "Question 1", "Question 2", "Question 3")

cat("=== SURVEY RESULTS TABLE ===\n")
print(results_table, row.names = FALSE)
```

```
=== SURVEY RESULTS TABLE ===
> print(results_table, row.names = FALSE)
  Respondent Question 1 Question 2 Question 3
          1          A          B          C
          2          B          A          D
          3          C          A          B
>
```

Output 3: Survey Results Table

Observation: The table shows each respondent's answers to all three questions with clearly labelled columns. Respondents 2 and 3 gave the same answer (A) for Question 2, while all other answers differ between respondents.

PROGRAM 4: Interactive Dashboard for Survey Results

Develop a Tableau dashboard combining the stacked bar chart, radar chart, and the table for interactive exploration of survey responses.

Note: Tableau could not be installed on the system used for this experiment, so the interactive dashboard was developed in R using the Shiny package, with the same charts and interactive filters required in the question.

```
library(shiny)

# Dataset for Set 14
survey_results <- data.frame(
  Respondent = c(1, 2, 3),
  Question_1 = c("A", "B", "C"),
  Question_2 = c("B", "A", "A"),
  Question_3 = c("C", "D", "B")
)
options <- c("A", "B", "C", "D")
q_cols <- c("Question 1" = "steelblue", "Question 2" = "darkorange", "Question 3" = "seagreen")

ui <- fluidPage(
  titlePanel("Survey Results Dashboard"),
  sidebarLayout(
    sidebarPanel(width = 3,
      checkboxGroupInput("qs", "Questions on radar chart:",
        choices = names(q_cols), selected = names(q_cols))
    ),
    mainPanel(width = 9,
      fluidRow(
        column(6, plotOutput("stackPlot", height = "300px")),
        column(6, plotOutput("radarPlot", height = "300px"))
      ),
      h4("Survey Results Table"),
      tableOutput("resultsTable")
    )
  )
)

server <- function(input, output) {
  output$stackPlot <- renderPlot({
    q1 <- table(factor(survey_results$Question_1, levels = options))
    barplot(matrix(q1, ncol = 1, dimnames = list(options, "Question 1")),
      col = c("steelblue", "darkorange", "seagreen", "orchid"), width = 0.5,
      xlim = c(0, 1.5), ylim = c(0, 3.5), ylab = "Number of Responses",
      main = "Answers for Question 1", legend.text = TRUE,
      args.legend = list(title = "Answer", x = "right", bty = "n"))
  }, width = 420, height = 300, res = 96)

  output$radarPlot <- renderPlot({
    angles <- pi / 2 - 2 * pi * (0:3) / 4
    par(mar = c(1, 1, 3, 1))
    plot(NA, xlim = c(-1.4, 1.4), ylim = c(-1.3, 1.3), asp = 1, axes = FALSE,
      xlab = "", ylab = "", main = "Radar Chart of Response Patterns")
    for (r in 1:2) polygon(r / 2 * cos(angles), r / 2 * sin(angles), border = "gray70")
    segments(0, 0, cos(angles), sin(angles), col = "gray70")
    text(1.18 * cos(angles), 1.18 * sin(angles), labels = options, font = 2)
  })
}
```

```

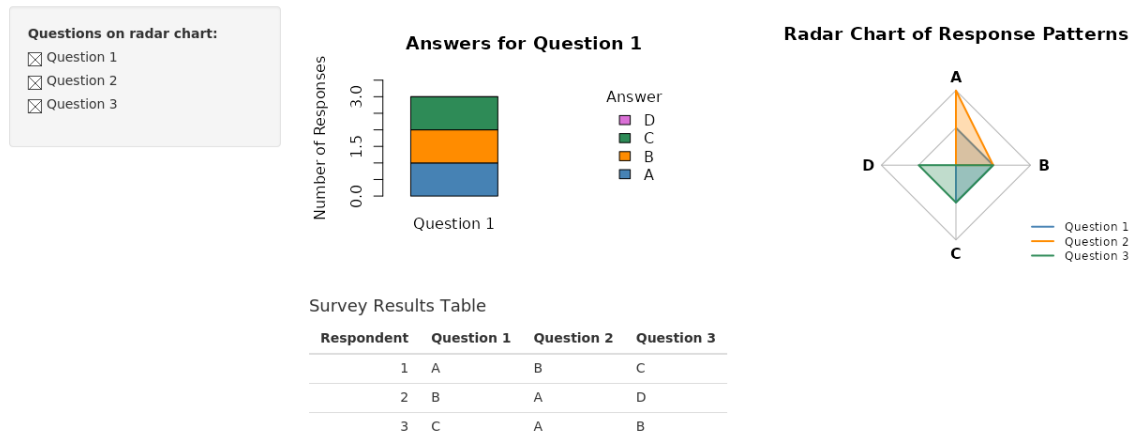
for (q in input$qs) {
  col_name <- gsub(" ", "_", q)
  r <- as.numeric(table(factor(survey_results[[col_name]], levels = options))) / 2
  polygon(r * cos(angles), r * sin(angles), col = adjustcolor(q_cols[q], 0.3),
    border = q_cols[q], lwd = 2)
}
legend("bottomright", legend = input$qs, col = q_cols[input$qs], lwd = 2, bty = "n", cex =
0.8)
}, width = 420, height = 300, res = 96)

output$resultsTable <- renderTable({
  d <- survey_results
  colnames(d) <- c("Respondent", "Question 1", "Question 2", "Question 3")
  d
}, digits = 0)
}

shinyApp(ui = ui, server = server)

```

Survey Results Dashboard



Output 4: Survey Results Interactive Dashboard (R Shiny)

Observation: The dashboard shows the Question 1 stacked bar chart, the radar chart, and the survey results table together. The question checkboxes choose which questions are drawn on the radar chart, so the response pattern of each question can be compared one at a time or together.

Procedure

1. Create the survey results data frame with Respondent and the answers to Questions 1, 2, and 3.
2. Count the Question 1 answers with `factor()` and `table()` and draw the stacked bar chart using `barplot()`.
3. Count the answers for every question, then draw the radar chart in base R using `polygon()`, `segments()`, and `text()`.
4. Rename the columns with `colnames()` and print the labelled results table.
5. Install and load the shiny package, and create the dashboard layout using `fluidPage()`, sidebar filters, and output areas for the charts and table.
6. Write the server function to filter the data with `reactive()` and redraw the outputs, then run the dashboard using `shinyApp()` and test the filters.

Result

The survey results were successfully visualized using a stacked bar chart, radar chart, and table in R, and an interactive R Shiny dashboard was built combining all three views.

Key R Functions Used

Function	Purpose
factor() / table()	Counts answers including options with zero responses.
barplot()	Creates the stacked bar chart.
rbind()	Combines the answer counts of all questions.
polygon()	Draws the radar grid and question shapes.
segments() / text()	Draws and labels the radar axes.
adjustcolor()	Adds transparency to the radar shapes.
colnames() / print()	Labels and displays the results table.
fluidPage() / checkboxGroupInput()	Creates the dashboard layout and filters.
reactive() / renderPlot()	Filters the data and redraws the charts.
renderTable()	Displays the filtered table on the dashboard.
shinyApp()	Runs the interactive dashboard.